

ORION: Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses

Smart India Hackathon 2026 | Project Report

Field	Details
Problem Statement ID	SIH26183
Organization	Ministry of Home Affairs
Department	Indian Cyber Crime Coordination Centre (I4C), CIS Division
Theme	Blockchain & Cybersecurity
Category	Software
Team	ORION

1. Executive Summary

Cyber-fraud victims (investment scams, task-based fraud, sextortion, ransomware, phishing, darknet payments) increasingly report the cryptocurrency wallet addresses that fraudsters used to collect money. Investigators then need to know **which exchange or Virtual Asset Service Provider (VASP) received the funds**, because only that entity can freeze assets and hand over KYC records.

Today this is done by manual blockchain tracing. It needs specialist skill, takes days, and gets harder with multi-chain transfers, bridges, mixers and DeFi protocols. By the time the exchange is found, the funds are usually gone.

ORION is a real-time crypto fraud attribution platform. It ingests a reported wallet address, builds a multi-chain transaction graph, detects laundering intermediaries, clusters exchange wallets, and identifies the **nearest exchange/VASP receiving a direct deposit**, with a confidence score. It then generates alerts, a freeze-request recommendation, and a standardised investigation report.

2. Problem Understanding

2.1 Background

Reported wallets are commonly:

- non-custodial wallets controlled by the fraudster,
- temporary burner wallets used once and abandoned,
- intermediary wallets used for layering and laundering.

2.2 Why the current process fails

Issue	Consequence
Manual tracing needs expertise	Few officers can do it; long queues
Multi-chain hops, bridges, DeFi swaps	Trail breaks at chain boundaries
Mixers/tumblers and privacy tools	Direct link between input and output is obscured
Exchange wallets are not labelled in one place	Officers cannot tell a deposit address from a private wallet
Delay in identifying the VASP	Late freezing, lost evidence, low recovery

2.3 What the problem statement asks for

1. Ingest wallet addresses from cybercrime complaint systems.

2. Automatically trace on-chain.
3. Identify associated exchanges/VASPs.
4. Detect fund-movement patterns.
5. Generate actionable intelligence for investigators.

Expected features: transaction graph analysis, exchange wallet clustering, intermediary wallet detection, cross-chain movement identification, SAHYOG/NCRP integration, automated alerts, wallet risk categorisation, multi-blockchain support, real-time tracing, investigative recommendations, dashboards, fund-flow visualisation, standardised reports, APIs, scalable indexing, and AI/ML-assisted typology recognition.

3. Proposed Solution

Pipeline: Complaint -> Wallet Trace -> Exchange Attribution -> Freeze and Recover

3.1 Modules

Module 1: Complaint ingestion

- Receives wallet address, chain, amount, timestamp and complaint ID via API (designed for NCRP/SAHYOG integration) or manual entry.
- Validates address format per chain and deduplicates repeat reports.

Module 2: Multi-chain indexing and tracing

- Pulls transaction data through blockchain node/indexer APIs.
- Follows funds forward hop by hop with configurable depth, minimum-value threshold and time window.
- Handles chain boundaries by matching bridge deposit and withdrawal events.

Module 3: Wallet transaction graph

- Entities: Wallet, Exchange/VASP, Transaction.
- Relationships: **Explicit** (on-chain transfer), **Inferred** (cluster link from heuristics), **Attributed** (deposit into a known exchange cluster).
- Stored in Neo4j; graph analytics via Graph Data Science.

Module 4: Laundering and intermediary wallet detection

Typology detectors:

1. Fan-in (many victims to one collector)
2. Fan-out / scatter
3. Layering (long chains of pass-through wallets)
4. Peel chain (small amounts peeled while the remainder moves on)
5. Mixer / bridge hop
6. Burner wallet (short-lived, one-time use)
7. Circular flow

Plus unsupervised OddBall egonet anomaly scoring and temporal burst scoring. Output: a **risk category** per wallet (for example collector, intermediary, mixer-linked, exchange deposit, unknown).

Module 5: Exchange/VASP attribution

- Clusters exchange wallets using common-input-ownership and deposit-address consolidation heuristics, combined with label data from blockchain intelligence APIs.
- Finds the first exchange cluster reached along each traced path and reports it as the **nearest exchange/VASP receiving a direct deposit**.
- Every attribution carries a **confidence score** and the supporting transactions.

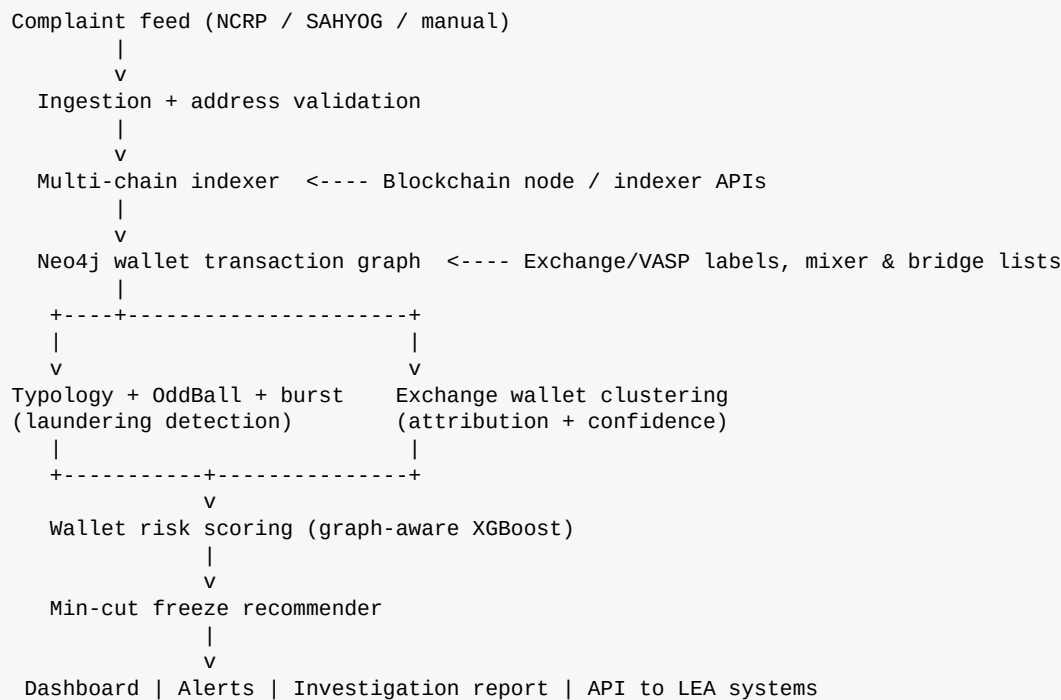
Module 6: Actions and reporting

- Real-time alerts (SMS/email/API) to investigators and I4C.
- Min-cut recommender: the smallest set of wallets/deposits where a freeze blocks the most traced funds.
- Auto-generated standardised investigation report (PDF) with the graph, timeline, attributed VASP, confidence and source transaction hashes.
- Draft freeze/preservation request text for the identified VASP.

3.2 Key design choices

- **Confidence over certainty.** On-chain attribution is probabilistic. ORION states confidence and evidence instead of asserting ownership.
- **Human in the loop.** No freeze request is sent without officer review.
- **Evidence-ready.** Every alert stores its source transaction hashes, timestamps and data source, supporting admissibility of electronic records.

4. System Architecture



5. Technology Stack

Layer	Technology
Backend and alerts	Python, FastAPI, message queue for real-time jobs
Graph storage and analytics	Neo4j, Graph Data Science (GDS), Cypher typology rules
Machine learning	scikit-learn, XGBoost (graph-aware features), OddBall
Blockchain data	Node/indexer APIs per chain (for example Bitcoin, Ethereum and EVM chains, Tron)
Dashboard	React, Cytoscape.js or D3 for fund-flow graphs, Tailwind CSS
Reports	Server-side PDF generation
Security	Role-based access, audit logs, encrypted storage

6. Data Sources

Source	Purpose
NCRP / SAHYOG complaints	Reported wallet addresses and case metadata
Blockchain node/indexer APIs	Transactions across supported chains
Exchange/VASP wallet label data	Ground truth for attribution
Mixer, bridge and threat-intel address lists	Flag obfuscation hops
Public labelled datasets (for example Elliptic)	Model training and validation
Simulated scam typologies	Testing investment, task-fraud and sextortion flows

7. Methodology

7.1 Tracing

1. Start from the reported address.
2. Expand outgoing transfers up to depth d and above value threshold v .
3. Collapse pass-through wallets (single input, single output) into chains.
4. At a bridge or swap, follow the matching event to the destination chain.
5. Stop a path when it reaches an exchange cluster, a mixer, or the depth limit.

7.2 Attribution

1. Group addresses likely controlled by the same entity (co-spend and deposit-consolidation heuristics).
2. Match clusters against exchange/VASP label sources.
3. Score confidence from label source quality, number of independent signals and cluster size.
4. Report the nearest exchange per path, ranked by confidence and by value received.

7.3 Model ladder (each stage must beat the one below)

1. Heuristic rules only
2. Rules plus clustering
3. Rules plus clustering plus graph-aware XGBoost risk score

Evaluation uses time-based splits, not random splits, to avoid the model memorising its own data.

7.4 Evaluation metrics

Metric	Meaning
Attribution precision / recall	Correct exchange identified vs. missed
Top-k attribution accuracy	Correct exchange among top k candidates
Trace time per case	Seconds from complaint to attribution
Wallet risk classification F1	Intermediary/collector detection quality
Alerts per day and precision	Investigator workload

8. Feasibility, Risks and Mitigation

Challenge / risk	Level	Mitigation
No labelled Indian fraud-wallet ground truth	HIGH	Public labelled datasets plus simulated scam typologies; production ingests NCRP/SAHYOG data via I4C

Challenge / risk	Level	Mitigation
Model memorises its own synthetic data	HIGH	Time-based splits, multiple scenarios, mandatory baseline, precision@K
Wrong attribution of exchange or owner	HIGH	Confidence score on every attribution, stored evidence, mandatory human verification before any freeze request
Mixers, bridges, privacy coins, chain-hopping	MEDIUM	Bridge event matching, adaptive adversary module, low-confidence flag and manual review where tracing is limited
Alert fatigue and false positives	MEDIUM	Tunable thresholds, explained alerts, reported precision and alerts per day
API cost and rate limits of data providers	MEDIUM	Caching, incremental indexing, pluggable provider layer
Exchange labels become outdated	MEDIUM	Periodic label refresh, provenance stored per label

Known limitation: transactions through strong privacy tools (for example privacy coins and some mixers) cannot always be traced. ORION flags such cases instead of guessing.

9. Impact and Benefits

Target audience

- State police and district cyber cells
- I4C, NCRP and SAHYOG teams
- Exchanges/VASPs and banks (faster, better-evidenced freeze requests)
- Citizens and victims (better recovery odds)

Key impact

Today	With ORION
Manual tracing needs expertise and days	Automated tracing to the nearest exchange/VASP in real time
Single-chain explorer lookups	Multi-chain, cross-bridge fund-flow attribution
Ad hoc notes as evidence	Standardised investigation report with source transactions
Late freeze requests	Freeze recommendation within the golden hour

10. Legal, Ethical and Privacy Considerations

- **Evidence integrity:** every alert stores source transaction hashes, timestamps and provider, supporting the admissibility of electronic records (BSA 2023) and the legal basis for freezing suspect funds (BNSS 2023).
- **Data protection:** lawful data handling, retention limits and role-based access aligned with the DPDP Act 2023.
- **No autonomous action:** ORION recommends; an officer decides.
- **Bias and error:** attribution is probabilistic and shown with confidence. A wallet flagged as suspicious is a lead, not a finding of guilt.

11. Future Scope

- Direct integration with SAHYOG and NCRP once access is granted.
- Standard request templates and secure channels to registered VASPs.
- Support for additional chains and layer-2 networks.
- Stablecoin-focused tracing (a common vehicle in scam payouts).

- Cross-case linking: recognising the same collector network across many complaints.
- Federated learning across state cyber cells without sharing raw data.

12. References

Academic

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2. Meiklejohn et al. (2013). "A Fistful of Bitcoins: Characterizing Payments Among Men with No Names." IMC.
3. Weber et al. (2019). "Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics." (Elliptic dataset)
4. Möser, Soska, Heilman et al. (2018). "An Empirical Analysis of Traceability in the Monero Blockchain." PoPETs.

Legal, industry and regulatory

5. I4C / NCRP / SAHYOG: complaint data and the fund-freezing workflow ORION plugs into.
6. FATF guidance on virtual assets and VASPs: laundering typologies underlying our detectors.
7. PMLA 2002 as applied to virtual digital asset service providers.
8. BNSS 2023 and BSA 2023: freezing of suspect funds and admissibility of electronic records.
9. DPDP Act 2023: data handling, retention and access control.